

Exploring Dynamics of WhatsApp Communications: Analyzing User Interactions and Behavior across Social and Professional Networks

***Prof. Pravin Patil^{1*}, Dhanashri Chaudhari², Jagruti Wagh³,
Divya Ahire⁴, Mohit Pawar⁵**

¹Assistant Professor, Department of Information Technology, SVKM's Institute of Technology, Dhule, Maharashtra, India

patilpravin555@gmail.com

²UG Student, Department of Information Technology, SVKM's Institute of Technology, Dhule, Maharashtra, India

chaudharidhanashri2603@gmail.com

³UG Student, Department of Information Technology, SVKM's Institute of Technology, Dhule, Maharashtra, India

jagrutiwagh2320@gmail.com

⁴UG Student, Department of Information Technology, SVKM's Institute of Technology, Dhule, Maharashtra, India

divyaahire3012@gmail.com

⁵UG Student, Department of Information Technology, SVKM's Institute of Technology, Dhule, Maharashtra, India

mohitpawar0503@gmail.com

Abstract

WhatsApp has emerged as a ubiquitous and efficient communication tool, widely used for both individual and group interactions, including voice and video calls. This study aims to analyze WhatsApp conversations to extract meaningful insights from interactions with friends, students, teachers, clients, and various groups. By leveraging Python libraries such as Pandas, Streamlit, Matplotlib, and SciPy, we perform a comprehensive analysis of WhatsApp chat data. The objectives include identifying the most active users, the frequency of messages sent by individuals, and the most common words used in group conversations. This analysis provides a deeper understanding of communication patterns and user behaviour within different social and professional contexts.

Keywords:

Chat analysis, Insight, WhatsApp, Python

1. Introduction

In the digital era, various chat applications facilitate messaging and communication, with WhatsApp being one of the most popular and widely used platforms. Over 65 billion messages are exchanged daily on WhatsApp, and users spend an average of 195 minutes per week on the platform, often participating in numerous group conversations. Given the vast amount of data generated, extracting meaningful insights from WhatsApp chats is crucial. This research work focuses on WhatsApp chat analysis using Python, leveraging its powerful tools for data handling, natural language understanding, and visual presentation of findings. By utilizing Python libraries such as Streamlit, Pandas, Matplotlib, and SciPy, we aim to analyze WhatsApp conversations to identify key patterns and insights. Streamlit, an open-source Python toolkit, plays a vital role in creating visually appealing and interactive online applications for data analysis. It allows for the development of complex data apps that facilitate an in-depth examination of WhatsApp chat data. Pandas, another essential library, is designed for data manipulation and provides data structures like Data Frame and Series, along with a wide range of functions for performing data operations. Our approach begins with extracting data from WhatsApp exports and ensuring it is clean and ready for analysis.[1] We then conduct exploratory data analysis to uncover insights such as message frequency, common topics, and user activity levels. The results of this analysis are presented through various graphs and visualizations, providing a comprehensive understanding of communication patterns and user behaviour within WhatsApp conversations.

2. Literature Survey

In today's internet-driven world, WhatsApp plays a crucial role in communication. This messaging app achieved the significant milestone of 200 million monthly active users just four years after its launch. Various studies have focused on analysing WhatsApp data to extract meaningful insights and understand user behavior. Ranjan, Vanishka Kapoor[1], and Deepika Bansal proposed a system for analysing data from exported WhatsApp chats over a 24-hour period. Their research utilized basic Python concepts for data analysis, presenting the results through clear and understandable visualizations. [2]Srishti Gulecha R, Muthu Reshmi K, and Dr. Abirami S developed a system to analyze WhatsApp chats for sentiment classification, identifying positive, negative, and neutral sentiments. Their approach involved behavior analysis of the top five users in a particular group or individual chat. This was achieved using machine learning techniques, including a linear Support Vector Classifier (SVC), to train and test the dataset for visualizing insights from [3]WhatsApp conversations. S. Jansi Rani, T. N. Prabhu, and J. Angel Ida Chellam[4] proposed a system using R for analysing WhatsApp data. Their work focused on visualizing data available in a 24-hour format to identify the most common user, the most active day, and the most active year. Additionally, their study visualized seven emotions: joy, anger, sadness, happiness, guilt, fear, and shame. These studies highlight the potential of leveraging data analysis techniques and tools to gain insights from WhatsApp chats, demonstrating the importance of this research area in understanding communication patterns and user behaviour.

3. METHODOLOGY

WhatsApp chat analysis aims to uncover the essence of group chats or personal conversations. This process involves three stages: exporting the data, data preprocessing, and data analysis. By utilizing these stages, we can systematically analyze WhatsApp data.[6][7][8]

Stage 1: Exporting the Data

In the first stage, the data is exported from WhatsApp. This involves using WhatsApp's built-in export feature to generate chat logs, which are then saved in a suitable format for further processing, typically as a text file.

Stage 2: Data Preprocessing

During the preprocessing stage, the extracted chat logs are converted into a structured format, typically a DataFrame, using the Pandas library. This involves parsing the raw text data to separate timestamps, sender names, and message content. The data is then cleaned to remove any irrelevant information or formatting issues, ensuring it is in a consistent and analysable format.

Stage 3: Data Analysis

Once the data is pre-processed, the analysis stage begins. Various techniques are applied to extract meaningful insights, including:

- **Message Frequency Analysis:** Counting the number of messages sent by each user to determine the most active participants.
- **Topic Modeling:** Identifying common topics or themes within the conversations.
- **Sentiment Analysis:** Classifying messages as positive, negative, or neutral to gauge the overall sentiment of the chat.
- **User Behavior Analysis:** Examining the behavior of individual users or groups, such as response times and interaction patterns.[9][10]

The results of the analysis are then visualized using tools like Matplotlib and Streamlit to create interactive and easily interpretable graphs and dashboards. These visualizations help in understanding the underlying trends and patterns in WhatsApp conversations, providing valuable insights into user interactions and communication dynamics.

A. Exporting the Data

To export WhatsApp chat data, follow these steps:

1. Open the WhatsApp application and select the group or personal contact whose chat you want to export.
2. Tap the three dots in the top right corner of the WhatsApp interface.
3. Select the "More" option and then choose "Export Chat."
4. The chat will be exported in text format.

After exporting the chats, it is in the text format contains the date, users name and message. The chat text file is as follows

```

21/11/2022, 14:09 - Messages and calls are end-to-end encrypted. No one outside of
them. Tap to learn more.
18/08/2022, 15:15 - Rohit Sir created group "SY IT AY2022-23"
21/11/2022, 13:18 - Rohit Sir: Class timing s 10:00 am
21/11/2022, 13:24 - Lokesh Sir: Roll number: 03, 12, 16, 41, 52 must submit C++ as
21/11/2022, 15:28 - Riddhi: <Media omitted>
21/11/2022, 15:29 - Riddhi: <Media omitted>
21/11/2022, 15:43 - Rohit Sir added +91 95112 46488
22/11/2022, 10:05 - 95112 46488: <Media omitted>
22/11/2022, 11:48 - Sagar Sir: Dear student CAO NPTEL result is announced, you are
information.

```

B. Preprocessing the Data

Before starting the project, ensure that Visual Studio Code and the necessary Python libraries (Streamlit, Matplotlib, Seaborn, Emojis, and URLExtract) are installed. First, create a web application using Streamlit. Next, add a file upload button in the sidebar of the web application to allow users to upload the chat text file for preprocessing.

```

st.sidebar.title("WhatsApp Chat Analyzer")
uploaded_file = st.sidebar.file_uploader("Choose a file")

```

The chat data is in text format, and for effective analysis, it must be converted into a structured format. We need to transform this text data into a Data Frame. This involves splitting the data into separate columns for the date, time, username, and messages. For example, a text entry like "21/11/2022, 14:09 - Rohit: Good morning" should be parsed into individual components: date, time, user, and message, to ensure the data is properly structured for analysis.

C. For Splitting the data write the Regular Expression

To split the data into separate components such as date, username, messages and time. Some of the people has the 12 hours clock in their device and some has 24 hours so the Regular expression is different for 12 hour and 24 hours.

The Regular Expression for 24 hours is:

```
pattern="\d{1,2}/\d{1,2}/\d{2,4}, \s\d{1,2}:\d{2}\s-\s"
```

The Regular Expression for 12 hours is:

```
patterns="\d{1,2}\d{1,2}\d{2,4}, \d{1,2}:\d{2}\s*[ap]m"
```

Splitting time and messages of the chats

```

f=open("chat.txt", 'r', encoding='utf-8')
data = f.read()
pattern="\d{1,2}/\d{1,2}/\d{2,4}, \s\d{1,2}:\d{2}\s-\s"

messages = re.split(pattern, data)[1:]
dates = re.findall(pattern, data)

```

	DATES	MESSAGES
0	21/01/2024, 14:22	Aakash: online section will be start in 10 minutes.

The above code, we split the data in the two parts time and messages using Regular Expression tailored for 24 hours.

```
f=open("chat.txt",'r',encoding='utf-8')
data = f.read()
patterns="\d{1,2}\d{1,2}\d{2,4}, \d{1,2}:\d{2}\s*[ap]m"

messages = re.split(pattern, data)[1:]
dates = re.findall(pattern, data)
```

	DATES	MESSAGES
0	16/06/2023, 1:14 pm	Aakash: online section will be start in 10 minutes.

The above code, we split the data in the two parts time and messages using Regular Expression tailored for 12 hours. This allows us to effectively analyses the data.

Separating chat messages from date and time

```
df=pd.DataFrame({"user_message":messages,"user_dates":dates})
df["user_dates"]=pd.to_datetime(df['user_date'],
    format='%d/%m/%Y, %H:%M - ')
df.rename(columns={'user_dates':'dates'},inplace=True)
```

	user_message	dates
0	Messages and calls are end-to-end encrypted. N...	2024-01-21 14:22:00
1	~Aakash saini created group "Learning and Int...	2024-01-20 14:55:00
2	You joined using this group's invite link\n	2024-01-21 14:22:00
3	Vaishnavi Ahirrao It joined using this group's...	2024-01-21 15:06:00
4	Divya joined using this group's invite link\n	2024-01-21 15:09:00

Figure 1: Splitting of chat messages for 24 hours clock

The above code demonstrates the splitting of the chat messages and the dates both for 24 hours format into distinct components.

```
df=pd.DataFrame({"user_message":messages,"user_dates":dates})
df["user_dates"] = pd.to_datetime(df['user_dates'],
format='%Y, %I:%M %p')
df.rename(columns={'user_dates':'dates'},inplace=True)
```

	user_message	dates
0	- Messages and calls are end-to-end encrypted...	2023-01-01 18:43:00
1	- Dr. Sunil Mane created group "IDE Bootcamp ...	2023-01-01 13:14:00
2	- You joined using this group's invite link\n...	2023-01-01 18:43:00
3	- Jayesh: Hello Everyone\n18/06/	2023-01-01 12:05:00
4	- Nikhil: At what time do we need to Report o...	2023-01-01 12:06:00

Figure 2: Splitting of chat messages for 12 hours clock

The above code demonstrates the splitting of the chat messages and the dates both for 12 hours format into distinct components

Splitting the username from the chat messages columns

```
user_list = []
message_list = []

for message in df['user_message']:
    entry = re.split('([\w\W]+?):\s', message)
    if entry[1:]:
        user_list.append(entry[1])
        message_list.append(entry[2])
    else:
        user_list.append("group_notification")
        message_list.append(entry[0])
df['user'] = user_list
df['message'] = message_list
df.drop(columns=['user_message'], inplace=True)
df.head()
```

	dates	users	messages
0	2023-01-01 18:43:00	Rohit Sir	Let's start the Online lecture
1	2023-01-01 13:14:00	Milind	Tomorrow students should bring their project
2	2023-01-01 18:43:00	+91 9458751236	Best luck for tomorrow
3	2023-01-01 12:05:00	- Jayesh	Hello everyone
4	2023-01-01 12:06:00	- Nikhil	At what time do we need to Report on 21st June...

Figure 3: Splitting of username

With the help above code the splitting of the dates, users and messages is done and can be identify quickly. Additionally, Data Frame functionalities, can be used to parse the dates column to split the time, year, month and date components. This structured format of the data will be used to quickly identify and analyse the data.

C. Displaying the data

After the preprocessing step the data is in the Structured Format and can be used to analyse and displaying the insights of the chats. It is done by using python libraries to extract the following insights:[11][12]

- Displaying number of Total Messages, Total words, Media used, Links shared in the WhatsApp Group
- Displaying Monthly timeline
- Displaying Daily Timeline
- Displaying Most Busy Day
- Displaying Most Busy Month
- Displaying Most Busy User
- Displaying Word Cloud
- Displaying Emojis Analysis

This will be displayed by using Matplotlib, Streamlit, and seaborn in the form of graphical representation.

4. RESULT/OUTPUT ANALYSIS

The Following is the analysis of the class group where all the study material, class related notice is shared. In this group there are both students as well as teachers. Let's analyse this group to find the insights of the groups.

Output of the Top Statistics

Total Messages	Total Words	Media Shared	Links Shared
1880	24520	552	126

Figure 4: Top Statistics

The figure 4 is showing the number of messages, words, media and links overall shared in the group.

Monthly Timeline

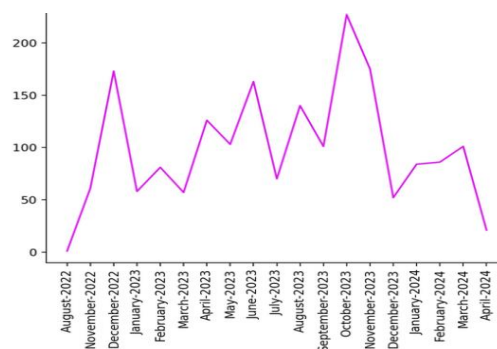


Figure 5: Monthly Timeline

The figure 5 is showing the Monthly Timeline of the group chat. It is starting from August 2022 to April 2024. The graph is the timeline of almost two years. It is also be seen that the busiest month of this graph is October 2023. In this way the Monthly timeline is represented.

Daily Timeline

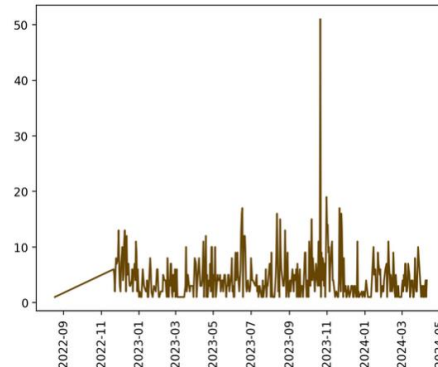


Figure 6: Daily Timeline

The above graph depicts the daily timeline of the group chat spanning from 2022-09 to 2024-05, covering nearly data of two years. Notably, 11 is the most active date in 2023 based on data presentation.

Most Busy Day

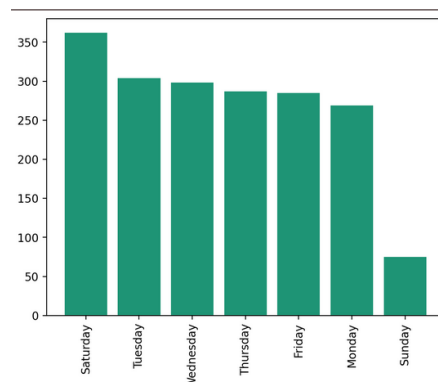


Figure 7: Most Busy Day

In figure7 illustrates the Busiest day for the group chat. According to the group chat on Saturday the highest number of messages are exchanged as determined through the overall analysis of the group chat. Basically, it highlights the Day of the Week when the group member is most active at communication. Top of Form

Most Busy Month

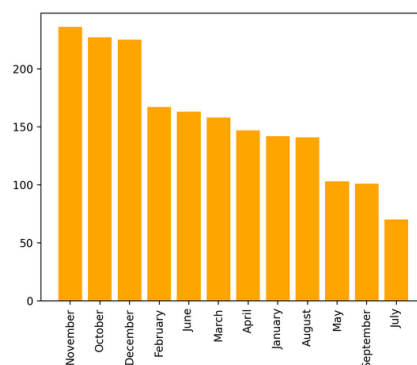


Figure 8: Most busy Month

The figure8 is the representation of the busiest month. November is the busiest month of the group chat. This is the analysis of the overall group but it can be done for Individual user also.

Weekly Activity Map

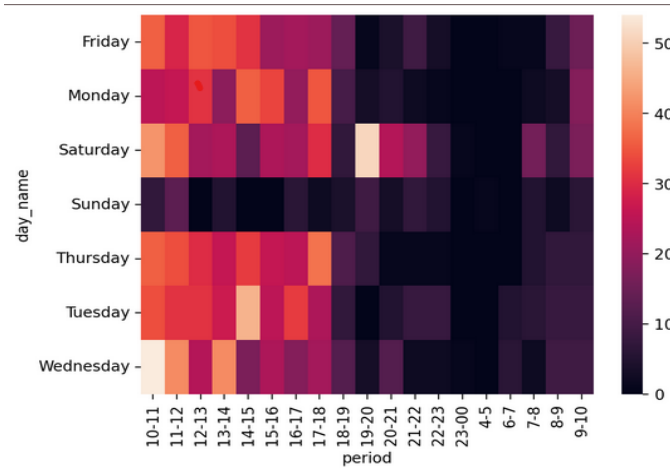


Figure 9: Weekly Activity Map

Above figure 9 is the heat map for the representation of the Weekly Activity. In heat map the number of messages is shown in the form of colours. When the number of messages gets increase the colour get lighter. It provides insights into specific date and time when message is most frequently sent within the group chat.

Most Busy User

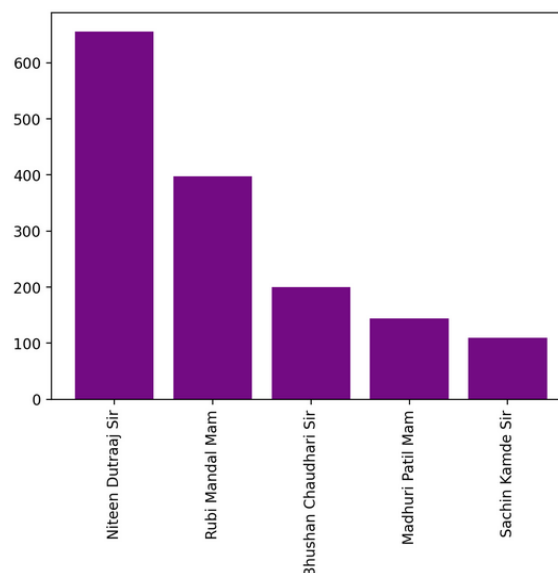


Figure 10: Most busy User

Figure 10 show the most active 5 users in the group. This is done by individually making count of the messages send by the user.

Figure 13 shows the Number of emojis mostly used in the group along with the pie chart representation

5. CONCLUSION

WhatsApp chat analysis can be effectively performed using Python and its libraries. Python's robust data processing capabilities allow for thorough analysis of exported chat data. The results of this analysis include metrics such as the total number of messages, words, media used, and links shared in WhatsApp groups. Additionally, insights can be gained from the monthly and daily timelines, identifying the most active days, months, and users. Visualizations such as word clouds and emoji analysis further enhance the understanding of communication patterns. This study demonstrates the potential of Python tools in extracting meaningful insights from WhatsApp data, with ample opportunities for further research and deeper analysis.

6. References

- [1] *Ranjan, Vanishka Kapoor, Deepika Bansal.* "Analysing WhatsApp Chat Using Python Libraries", "Maharaja Agrasen Institute of Technology, Delhi" 2022 3rd International Conference for Emerging Technology (INCET)
- [2] *Srishti Gulecha R, Muthu Reshmi K, Dr. Abirami S,* "Exploratory Data Analysis of WhatsApp group chat", "Department of Information Science and Technology College of Engineering, Guindy Chennai, India" 2023 12th International Conference on Advanced Computing (ICoAC)
- [3] *Dr.T.Siva Ratna Sai1, T.Naga Nandini2, M.Harsha Vardhan3, B.Sivaji4, R.Sai Kalyan,* "Chat Analysis on Whatsapp Using Machine Learning", "Department of Computer Science and Engineering, Qis college of Engineering and Technology, Ongole, Andhra Pradesh, India", Vol 14 Issue 04, 2023
- [4] *S.Jansi Rani, T.N.Prabhu, J.Angel Ida Chellam,* "Whatsapp Sentiment Analysis Using R", "Sri Ramakrishna Engineering College, Coimbatore", 2022 3rd International Conference for Emerging Technology (INCET) Belgaum, India. May 27-29, 2022
- [5] *S. Dahiya, A. Mohta and A. Jain,* "Text Classification based Behavioural Analysis of WhatsApp Chats," 2020 5th International Conference on Communication and Electronics Systems (ICCES), Coimbatore, India, 2020, pp. 717-724, doi: 10.1109/ICCES48766.2020.9137911.
- [6] *Ravishankara K, Dhanush, Vaisakh, and Srajan I S,* "Whatsapp Chat Analyzer," International Journal of Engineering Research and, vol. V9,no. 05, May 2020, doi: <https://doi.org/10.17577/ijertv9is050676>.
- [7] *M., Wattenberg and D. Fisher.* "Analyzing Perceptual Organization in
- [8] *Information Graphics," Information Visualization, vol 3, no.2, PP. 123-133., 2004, doi: 10.1057/palgrave.ivs.9500070*
- [9] *S. Shahid,* "Content Analysis of Whatsapp Conversations: An Analytical Study to Evaluate the Effectiveness of Whatsapp Application in Karachi," International Journal of Media, Journalism and Mass Communications, vol. 4, no. 1, 2018

- [10]K. Kelley, "What is data analysis? process, types, methods and Techniques," *Simplileam.com*, 3 1-Jan-2023.
- [11]Ahmed, I., Fiaz, T., "Mobile phone to youngsters: Necessity or addiction", *African Journal of Business Management* Vol.5 (32), pp. 12512-12519, Aijaz, K. (2011).
- [12]Jessica Ho, Ping Ji, Weifang Chen, Raymond Hsieh, "Identifying google talk", *IEEE International Conference on Intelligence and Security Informatics, ISI '09*, pp. 285-290, 2009.
- [13]Mike Dickson, "An examination into AOL instant messenger 5.5 contact identification.", *Digital Investigation, ScienceDirect*, vol. 3, issue 4, pp. 227-237, 2006
- [14]Aharony, N., T., G., The Importance of the WhatsApp Family Group: An Exploratory Analysis. "Aslib Journal of Information Management, Vol. 68, Issue 2, pp.1-37" (2016)
- [15]D.Radha, R. Jayaparvathy, D. Yamini, "Analysis on Social Media Addiction using Data Mining Technique", *International Journal of Computer Applications (0975 – 8887)* Volume 139 – No.7, pp. 23- 26, April 2016